

Database Systems and Laboratory

General Course Info (yes, again)

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Bachelor degrees in Internet of things, Big Data, Machine Learning, & Computer Science
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Who am I?



Nicola Saccomanno,
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- ▶ Feel free to reach out (multiple times) for any reason:
`nicola.saccomanno@uniud.it`
- ▶ Neurosymbolic AI, interpretability, machine and deep learning, wireless sensing and positioning, AI for healthcare, data science
- ▶ Personal website:
`http://users.dimi.uniud.it/~nicola.saccomanno`
- ▶ Data Science and Automatic Verification Lab:
`https://datasciencelab.dimi.uniud.it/`
- ▶ Our group is happy to support you throughout the development of your (bachelor or master) thesis! For "standard" project examples look here
`https://servizi.amm.uniud.it/CercaTesi/Home/Results/`
→ "Ricerca Avanzata" and type "ANGELO MONTANARI" in the "Nominativo relatore" field; for advanced project (AI, ML, FM, Logic, ...) see our publications at
`https://datasciencelab.dimi.uniud.it/` → "Publications"

Some details about the course

- ▶ Course title: Database Systems and Laboratory, Bachelor degrees in Internet of things, Big Data, Machine Learning, & Computer Science
- ▶ The course will be held in ITALIAN (both the lecture as well as the examination); the material (slides) will be English
- ▶ Course overview: overall, 12 CFU (9 CFU "theory" + 3 CFU lab)
 - ▶ First semester: 24 hrs (18 hrs + 6 lab hrs)
 - ▶ Second semester: 72 hrs (54 hrs + 18 lab hrs)
- ▶ Second semester schedule:
 - ▶ Theory lectures - Tuesday, Wednesday, Thursday 10.30 AM - 12.30 PM
 - ▶ Lab lectures - Tuesday 2.30 PM - 4.30 PM
 - ▶ Probably no lecture in some days at the end of April due to academic commitments
- ▶ Office hour (A2 51): Thursday 2.00 PM - 3.00 PM (prior confirmation via email)

Info about the exam

Examination (for 12 CFU): Written exam + oral exam + Lab project

- ▶ Written part: mandatory, 2-3hrs, 3-4 exercises on the course's topics (Conceptual modeling, Logical modeling, RA + SQL, DDL + Trigger, Transaction management). Grade range 0-30L (16+ to pass)
- ▶ Oral part: optional, upon student's request to improve the grade; mandatory (i.e., otherwise exam considered as failed) if written exam mark within $[16, 18[$, or if requested by the course's instructors
 - ▶ Oral exam involves exercises as well theoretical questions
 - ▶ Possibility to do the oral exam in a different session
- ▶ Lab project: according to dr. Geatti instructions (if you do not know them or you are not in a group for the lab project, write to luca.geatti@uniud.it ASAP!)

Some tips to pass the exam...

- ▶ Come to follow the lectures!
 - ▶ If you follow and understand the course, the exam will likely be passed with minimal effort
 - ▶ Possibility to interact and ask for questions/clarifications
 - ▶ On some days we will do exercises; there you can test live your understanding of the content of the course, while I'll provide feedback and discussions
- ▶ Try to do again by yourself the exercises we see during the lectures
- ▶ Ask for an appointment (office hour / ricevimento) to discuss the exercises if you have any doubt!

Material for the course

► Official communications and material available on Microsoft Teams:

- Subscribe (<https://goto.uniud.it/to/782>) and tell me ASAP if you have issues
- Please, double check that you are enrolled to the 2025-2026 Team

► Recommended:

- Either
 - **Basi di Dati**, McGraw-Hill. 5a (2018) o 6a (2023) Edizione. P. Atzeni, S. Ceri, P. Fraternali, S. Paraboschi and R. Torlone. — [ENG] **Database Systems - Concepts, Languages and Architectures**. McGraw-Hill, 1999. P. Atzeni, S. Ceri, S. Paraboschi and R. Torlone. Freely available <https://dbbook.inf.uniroma3.it/>.
 - **Fundamentals of Database Systems**. 7th Edition. Pearson International Education / Addison Wesley, 2016. R. Elmasri, and S. Navathe ([ITA] Sistemi di basi di dati. – 7/ED, Pearson)

► Teacher slides, exercises and code available on Teams.

► Recordings: lectures + exercises (from previous years by prof. Montanari)

► Optional:

- Advance, comprehensive book: Database System Concepts. McGraw-Hill, Seventh Edition. Avi Silberschatz, Henry F. Korth, S. Sudarshan.

A sketch (very high level) of the course program

- ▶ Introduction, the concept of abstraction, and overview of DB systems
- ▶ Conceptual modeling
- ▶ Logical modeling and normalization
- ▶ Relational Algebra
- ▶ Database creation
- ▶ Database manipulation (Structured Query Language and triggers)
- ▶ Concurrency and transaction management
- ▶ Physical layer
- ▶ Exercises for the exam

Feedback!

- ▶ Feedback at the end of the course is very useful to prepare next year edition of the course
- ▶ Feedback throughout the course is very appreciated, and useful to improve this year edition while doing it (anonymous form: <https://forms.gle/mCMMnrEvJW21epxL7>)!